

Perform the indicated operation and simplify.

6. $(4z^5 + 9z^4 + 17) - (-6z^4 + z^5 + 13)$

$$= 4z^5 + 9z^4 + 17 + 6z^4 - z^5 - 13$$

$$= \underline{3z^5 + 15z^4 + 4}$$

7. $(8x - 1)(x^3 + x^2 + x)$

$$= 8x(x^3 + x^2 + x) - 1(x^3 + x^2 + x)$$

$$= 8x^4 + 8x^3 + 8x^2 - x^3 - x^2 - x$$

$$= \underline{8x^4 + 7x^3 + 7x^2 - x}$$

Factor Completely.

8. $x^2 - x - 6$

$$\underline{(x-3)(x+2)}$$

$$6 = (-3)(2)$$

$$-3 + 2 = -1$$

9. $4x^3 - 4x^2 - 24x$

$$= 4x(x^2 - x - 6)$$

$$= \underline{4x(x-3)(x+2)}$$

10. $x^2 - 64$

$$= \underline{(x-8)(x+8)}$$